



Fig. 5: Get Servoblaster from git



Fig. 6: Change to the Servoblaster directory



Fig. 7: Process number



Fig. 8: Creating an entry for the device

for all eight servo motors, as shown in Fig. 4. The positive pin of each servo motor is connected with the positive pin of the external power supply and the ground pin of the power supply, the servo motors and Raspberry Pi are connected to each other to make a com-

mon ground.

How the project works is simple. Once you follow up the software section and install everything, you can control all servos independently. When you switch on the Raspberry Pi board, it will start booting up. The driver creates a device file '/dev/servoblaster' into which you can send commands. The command format is "<servo-number>=<servo-position>", where the

servo number is a number from 0 to 7 inclusive, and the servo position is the pulse width you want in units of 10 μ s. So, if you want to set Servo 3 to a pulse width of 1 ms, you could do this at the shell prompt as: 'echo 3=100 > /dev/servoblaster'. As a result, Servo motor

3 comes to the 0 degree position.

Software

Servoblaster is a Linux kernel driver for Raspberry Pi, which provides an interface to drive multiple servo motors via the GPIO pins without additional hardware such as microcontrollers. It seems to work very nicely, giving good stable outputs even when Raspberry Pi is busy with other tasks.

Assuming you have the Debian Wheezy Linux distribution on your Raspberry Pi, follow the steps given below: First, update and upgrade the Raspberry Pi related software and reboot your Raspberry Pi for a fresh look.

Once the system reboots, install 'gcc' for C compilation and 'git' to set up a repository or create an empty directory using the commands shown below:

```
sudo apt-get install gcc
sudo apt-get install git
```

Now, to get Servoblaster from git, type the following command in the terminal (refer to Fig. 5):

```
git clone git://github.com/
```



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